Customer Shopping Behavior Analysis

Project Overview

This project analyzes customer purchasing behavior to identify patterns that can improve sales performance, customer satisfaction, and long-term customer retention.
The objective is to understand how factors such as demographics, product categories, discounts, payment methods, and seasonal trends influence customer decisions.

Tools Used

- Python (Pandas) – Data cleaning & preprocessing
- PostgreSQL – Data analysis using SQL
- Power BI – Data visualization

Dataset Summary

The dataset contains customer transaction data used to analyze purchasing behavior.

- **Rows:** 3,900
- **Columns:** 18

Key Features:

- **Customer Demographics (** Age, Gender, Location, Subscription Status **)**
- **Purchase Details (** Item Purchased, Category, Purchase Amount **(USD))**
- **Shopping Behavior (** Previous Purchases, Frequency of Purchases **)**
- **Product Attributes (** Size, Color **)**
- **Marketing & Influence Factors (** Discount Applied, Review Rating **)**
- **Transaction Details (** Payment Method, Shipping Type **)**
- **Time Factor (** Season **)**

Missing Values:

Review Rating : 37 missing values
Handled using ***median imputation grouped by category***

- [Exploratory Data Analysis (Python)]

3.1 Data Loading

The dataset was loaded into Python using Pandas.

3.2 Initial Exploration

Initial inspection was performed using:

- df.head()
- df.info()
- df.describe()

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

3.3 Data Cleaning & Preprocessing

- Checked missing values using df.isnull().sum()

- Filled missing values in review_rating using category-wise median

- Standardized column names to lowercase and snake_case

- Removed redundant column promo_code_used

- **Created new features:**

- age_group (Young Adult, Adult, Middle-aged, Senior)
- Converted purchase frequency into numeric days

3.4 Data Integration

The cleaned dataset was connected to PostgreSQL using SQLAlchemy and psycopg2.

- ## Data Analysis (PostgreSQL)

4.1 Revenue Distribution by Gender

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Insight: Male customers generate significantly higher revenue.

4.2 Discount Impact on High-Spending Customers

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
Total rows: 839	Query complete 00:00:00.155	

Insight: Discounts are effective even for high-value customers.

4.3 Top Rated Products

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Insight: Footwear and accessories have higher satisfaction.

4.4 Shipping Type vs Spending

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Insight: Express shipping customers spend slightly more.

4.5 Subscription Analysis

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Insight: Subscription does not significantly increase spending.

4.6 Discount-Driven Products

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

Insight: These products are highly influenced by discounts.

4.7 Customer Segmentation

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Insight: Strong customer retention with a large loyal base.

4.8 Top Products by Category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Insight: Different categories have distinct top-selling products, showing varied customer preferences across product segments.

4.9 Repeat Buyers vs Subscription

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Insight: Repeat buyers are not necessarily subscribers.

4.10 Revenue by Age Group

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Insight: Young adults contribute the highest revenue.

4.11 Payment Method Analysis

	payment_method text	total_orders bigint	avg_spend numeric	total_revenue numeric
1	Credit Card	671	60.07	40310
2	PayPal	677	59.25	40109
3	Cash	670	59.70	40002
4	Debit Card	636	60.92	38742
5	Venmo	634	58.95	37374
6	Bank Transfer	612	59.71	36544

Insight: Credit cards generate the highest revenue, while debit cards show the highest average spending.

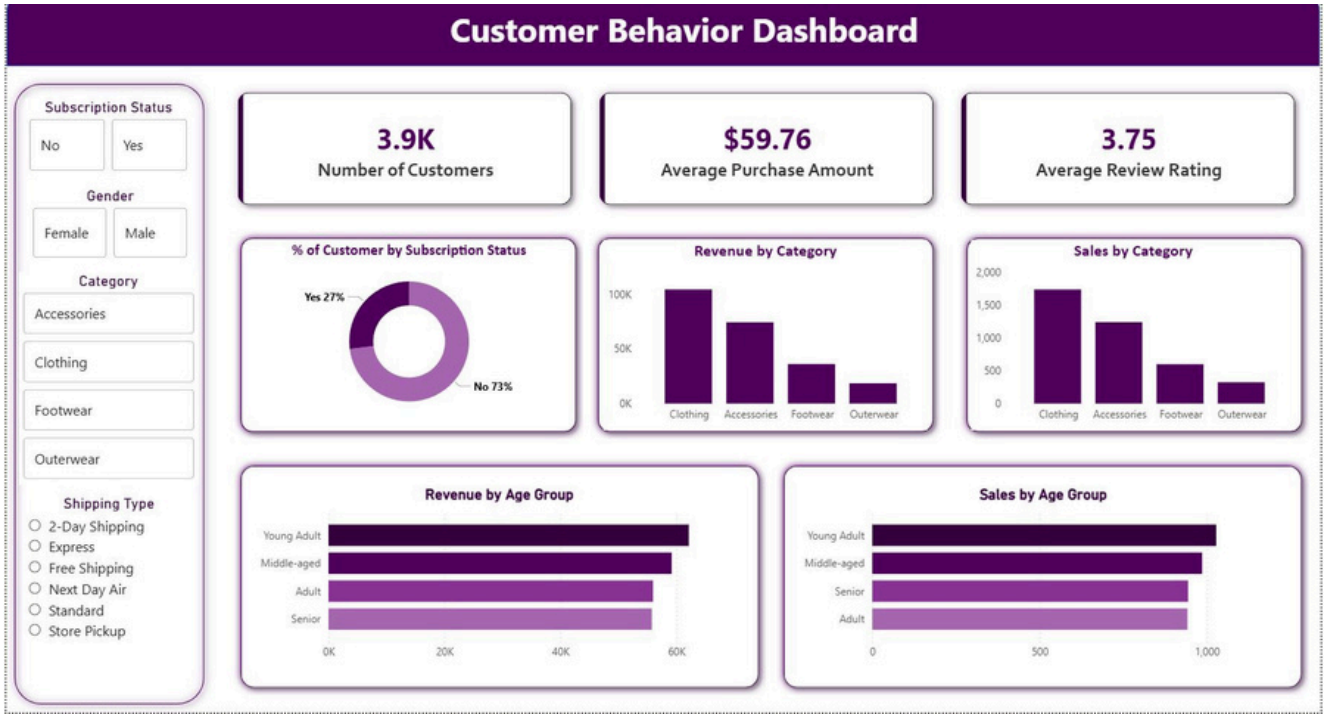
4.12 Seasonal Analysis

	season text	total_revenue numeric	avg_spend numeric
1	Fall	60018	61.56
2	Spring	58679	58.74
3	Winter	58607	60.36
4	Summer	55777	58.41

Insight: Fall performs best, while summer shows lower performance

- ## Dashboard (Power BI)

A Power BI dashboard was developed to present key metrics, including total customers, average purchase amount, and average review rating. It provides category-wise revenue and sales analysis, along with revenue and sales distribution across age groups. Interactive filters for subscription status, gender, season, and category enable dynamic data exploration and analysis.



- ## Business Recommendations

  - Target young adults as primary revenue drivers
  - Optimize discount strategies for maximum impact
  - Improve subscription model value
  - Focus on top-performing products per category
  - Use seasonal trends for marketing campaigns
  - Strengthen customer retention strategies

- ## Conclusion

This analysis highlights key factors influencing customer behavior, including demographics, product preferences, discounts, and seasonal trends.

The insights can help businesses make informed decisions to improve revenue and customer engagement.